RANYA RAJESH PRASAD

Embedded Software · C / C++ · Robot Firmware & Hardware Interfaces · AWS · AI-Assisted Development · Python

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EDUCATION

University of Illinois Urbana-Champaign — B.S. Electrical Engineering (ECE Department) **Expected May 2029**

Grainger College of Engineering • Sophomore, junior standing by credit hours • GPA **3.65**/4.00 • **Minor: Hoeft Technology & Management Program** (Gies College of Business and Grainger College of Engineering) • Available Summer 2027

Coursework: ECE 220 Computer Systems & Programming (C, C++, assembly, memory), **ECE 120 Introduction to Computing (digital logic, computer organization)**, ECE 110 Introduction to Electronics, ECE 210 Analog Signal Processing, CS 173 (Discrete Structures), MATH 241 (Multivariable Calculus), MATH 257 (Linear Algebra) • CS 225 Data Structures in C++ (Spring 2027)

RECOGNITION

Corresponding author, peer-reviewed international publication, "SkinSight: An App Leveraging Neural Networks for Skin Cancer Detection," *Intl. Journal of Pharma Medicine & Biological Sciences* (2025) · **Oral presentation, ICBCB 2025** — IEEE co-sponsored, Seoul National University · **First Place, shsHacks** (judged by Hudson River Trading) · Illinois State Scholar (top 10%)

TECHNICAL SKILLS

Embedded: C, C++, assembly, Arduino firmware, Raspberry Pi, sensor-driven control loops, serial communication (UART over USB, 9600 baud), PWM motor control, ultrasonic sensing, sensor filtering, fail-safe design

AI-assisted development: Claude Code, GitHub Copilot; I validate AI-generated code with review and tests before merge

Cloud & tools: AWS (Cognito, ECS/Fargate, CloudWatch), Terraform, Docker, Git, code review, GitHub Actions CI/CD, Linux

Test & troubleshooting: hardware simulator on the production code path, reference test suites, regression checks in CI, system monitoring and log analysis • **Also: Python**, Java, JavaScript / TypeScript, SQL, Linux, Docker, AWS, TensorFlow Lite

EMBEDDED & SYSTEMS PROJECTS

Software Chair — *WinnieWheels autonomous delivery robot* · Women in ECE, UIUC · C++, Arduino, Raspberry Pi **Oct 2025 – Sep 2026**

– **The first layer of code on the device**: the obstacle-avoidance loop (two HC-SR04 ultrasonic sensors in, an L298N motor driver out over PWM) runs in C++ on the Arduino, so the robot stays safe if Wi-Fi drops or the host stalls.

– **Stability**: one bad sensor reading made the robot brake for nothing. Moved the firmware to a **three-sample read with validity filtering** that rejects out-of-range and timed-out values; total sensor failure is a stop-and-retry.

– **I/O performance**: the navigation loop re-sent the same command every cycle and **saturated the serial link**. Added throttling and last-action deduplication so a command is sent only when the decision changes.

– **Hardware/software interface**: specified the newline-delimited JSON protocol between firmware and host and built the bridge that parses sensor frames and writes motor commands; wrote the motor algorithm and **coordinated integration with the hardware chairs**.

– **Test and troubleshoot without hardware**: built a simulator that feeds synthetic sensor frames through the identical code path. **Co-led software on a 50-person team**, ran code reviews, and ran the robot live at Engineering Open House; still a team member.

NetWatch — monitoring the stability of network devices · *Python, GitHub Actions, pytest* **2026**

– **System monitoring and analysis**: parses device logs and detects link flaps, lost adjacencies and login storms with statistical baselines plus rules. **Quality gate**: CI scores every change against a labeled test set and blocks it if accuracy drops.

SkinSight — a neural network running on the device · *Java (Android), TensorFlow Lite* · Lead & Corresponding Author **2023 – 2025**

– Compared four neural networks over three trials each (best: Transfer Learning, 84.69% mean accuracy, $p = 0.021$) and **ran the chosen model on the phone itself** through TensorFlow Lite, with no server. Published and presented internationally.

Parallel computing: genome variant-calling pipeline · *Linux cluster; bwa, samtools* · UChicago **Summer 2023**

– **Measured and improved system efficiency**: a head node dispatched jobs to worker nodes, tuned by worker count and threads per node; found the bottleneck stages and scaled from one chromosome (2 hours) to all 23 (16 hours) at 8× the cost.

EXPERIENCE

Jr. Applied AI Developer (IRAP) — *AbbVie* (contract through HireGenics) · Information Research **Oct 2026 – Present**

– **Software quality assurance**: building test frameworks that decide when AI output can be trusted, with regression detection in CI/CD.

Software Developer Intern — *Fylm TV, Inc.* · AWS, Terraform, Docker, CI/CD **Aug 2026 – Present**

– **AWS infrastructure**: own the Terraform module for the platform's authentication layer on AWS in a shared codebase, and hardened the checks that gate every pull request.

Software Developer — *NOBE* · client engagement for a registered investment advisory firm · Python, React **Sep 2026 – Present**

– Building financial features for a paying client, with a calculation engine **validated against the client's 13-scenario reference test suite**.

Also: Crisis Management App (2022), a Java Android app on Firebase (Camera, Location, SMS APIs) · Cafe Queues (2022–2026), a Flask and PostgreSQL backend tested with 20 simultaneous orders · EXPO Technical Lead & EOH Special Events Director, Engineering Council